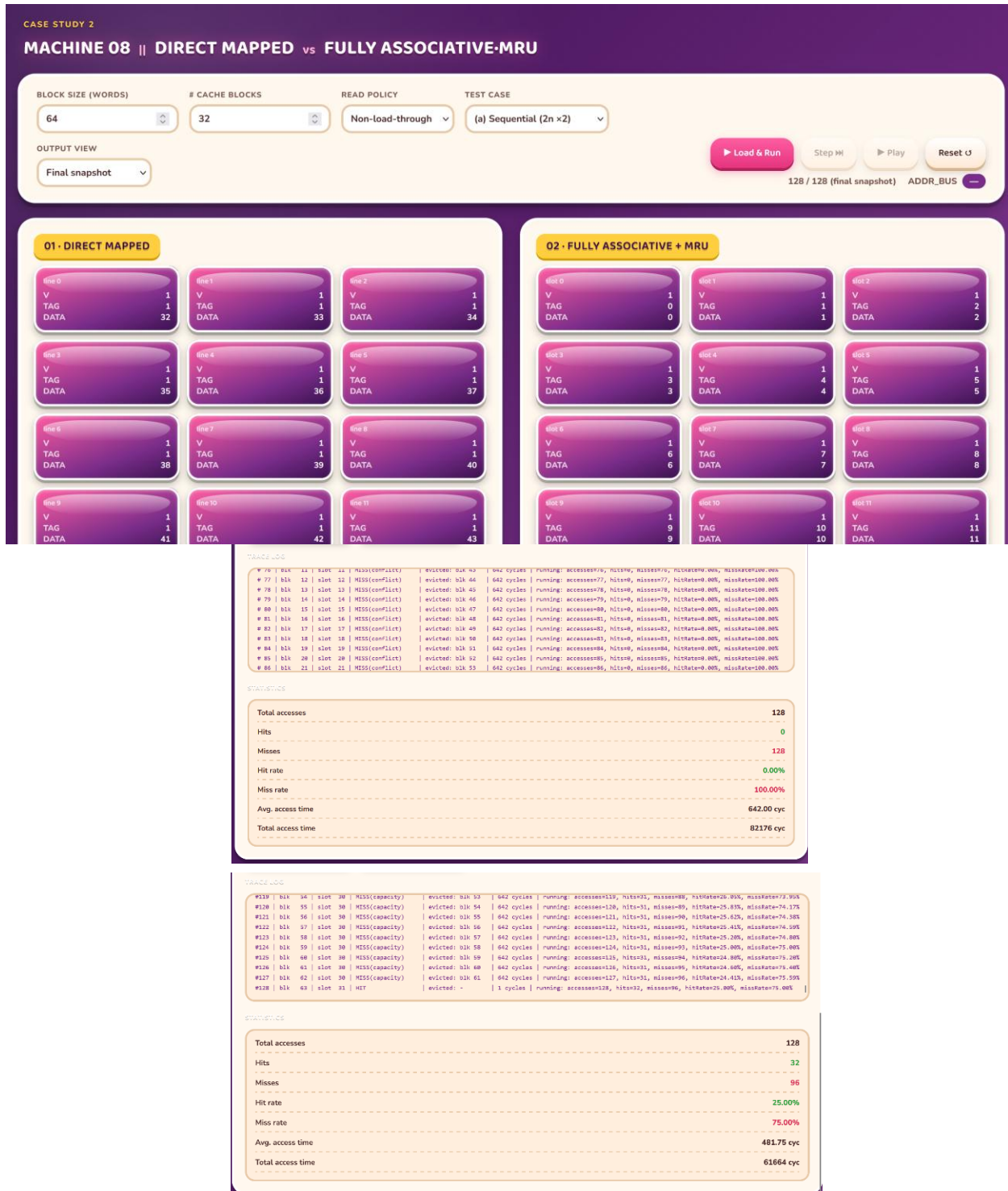
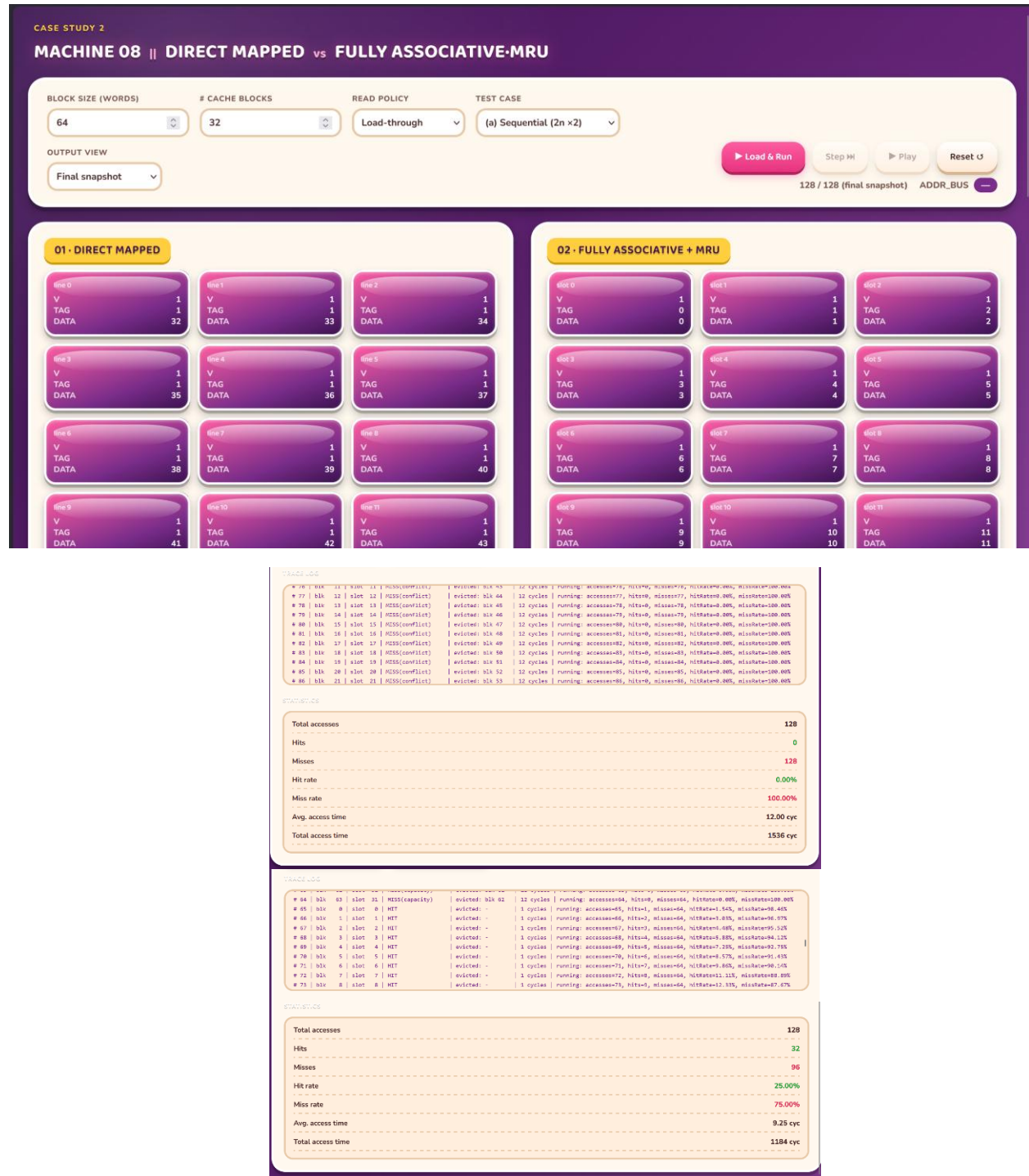


Test Case 1:

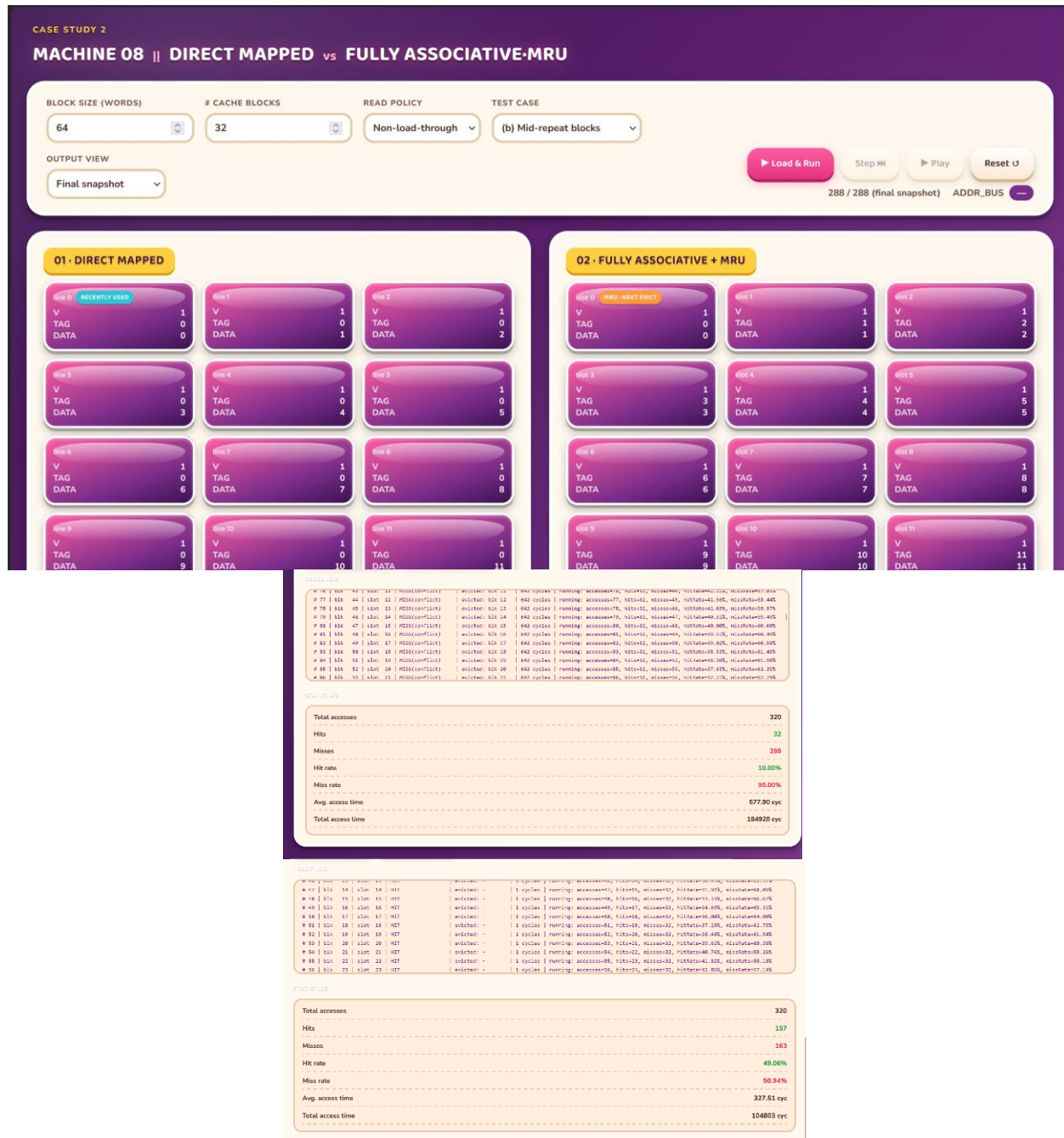
Non-Load Through



Load Through



Non-Load-Through



[illegible]

Test Case 3

Non-load through

CASE STUDY 2

MACHINE 08 || DIRECT MAPPED vs FULLY ASSOCIATIVE-MRU

BLOCK SIZE (WORDS): 64 # CACHE BLOCKS: 32 READ POLICY: Non-load-through TEST CASE: (c) Random (64 accesses)

OUTPUT VIEW: Final snapshot

64 / 64 (final snapshot) ADDR_BUS

01 - DIRECT MAPPED

line 0	line 1	line 2	line 3	line 4
V: 1 TAG: 5 DATA: 160	V: 1 TAG: 0 DATA: 1	V: 1 TAG: 8 DATA: 258	V: 0 TAG: — DATA: —	V: 1 TAG: 15 DATA: 484
line 5	line 6	line 7	line 8	line 9
V: 1 TAG: 22 DATA: 709	V: 1 TAG: 14 DATA: 454	V: 1 TAG: 11 DATA: 359	V: 1 TAG: 21 DATA: 680	V: 1 TAG: 1 DATA: 41
line 10	line 11	line 12	line 13	line 14
V: 1 TAG: 10 DATA: 330	V: 1 TAG: 203 DATA: 203	V: 1 TAG: 4 DATA: 140	V: 0 TAG: — DATA: —	V: 1 TAG: 19 DATA: 622
line 15	line 16	line 17	line 18	line 19
V: 1 TAG: 8 DATA: 271	V: 1 TAG: 27 DATA: 880	V: 1 TAG: 15 DATA: 497	V: 1 TAG: 26 DATA: 850	V: 1 TAG: 16 DATA: 531

02 - FULLY ASSOCIATIVE + MRU

slot 0	slot 1
V: 1 TAG: 585 DATA: 585	V: 1 TAG: 930 DATA: 930
slot 5	slot 6
V: 1 TAG: 778 DATA: 778	V: 1 TAG: 805 DATA: 805
slot 10	slot 11
V: 1 TAG: 454 DATA: 454	V: 1 TAG: 203 DATA: 203
slot 15	slot 16
V: 1 TAG: 258 DATA: 258	V: 1 TAG: 773 DATA: 773

1st iteration

TRACE LOG

```
--- Direct Mapped | non-load-through | blocksize=64 | numCacheBlocks=32 ---
# 1 | blk 380 | slot 9 | MISS(compulsory) | evicted: - | 642 cycles | running: accesses=1, hits=0, misses=1, hitRate=0.00%, missRate=100.00%
# 2 | blk 930 | slot 2 | MISS(compulsory) | evicted: - | 642 cycles | running: accesses=2, hits=0, misses=2, hitRate=0.00%, missRate=100.00%
# 3 | blk 41 | slot 8 | MISS(compulsory) | evicted: blk 380 | 642 cycles | running: accesses=3, hits=0, misses=3, hitRate=0.00%, missRate=100.00%
# 4 | blk 781 | slot 18 | MISS(compulsory) | evicted: - | 642 cycles | running: accesses=4, hits=0, misses=4, hitRate=0.00%, missRate=100.00%
# 5 | blk 838 | slot 22 | MISS(compulsory) | evicted: - | 642 cycles | running: accesses=5, hits=0, misses=5, hitRate=0.00%, missRate=100.00%
# 6 | blk 778 | slot 16 | MISS(compulsory) | evicted: - | 642 cycles | running: accesses=6, hits=0, misses=6, hitRate=0.00%, missRate=100.00%
# 7 | blk 885 | slot 5 | MISS(compulsory) | evicted: - | 642 cycles | running: accesses=7, hits=0, misses=7, hitRate=0.00%, missRate=100.00%
# 8 | blk 622 | slot 14 | MISS(compulsory) | evicted: - | 642 cycles | running: accesses=8, hits=0, misses=8, hitRate=0.00%, missRate=100.00%
# 9 | blk 355 | slot 31 | MISS(compulsory) | evicted: - | 642 cycles | running: accesses=9, hits=0, misses=9, hitRate=0.00%, missRate=100.00%
```

STATISTICS

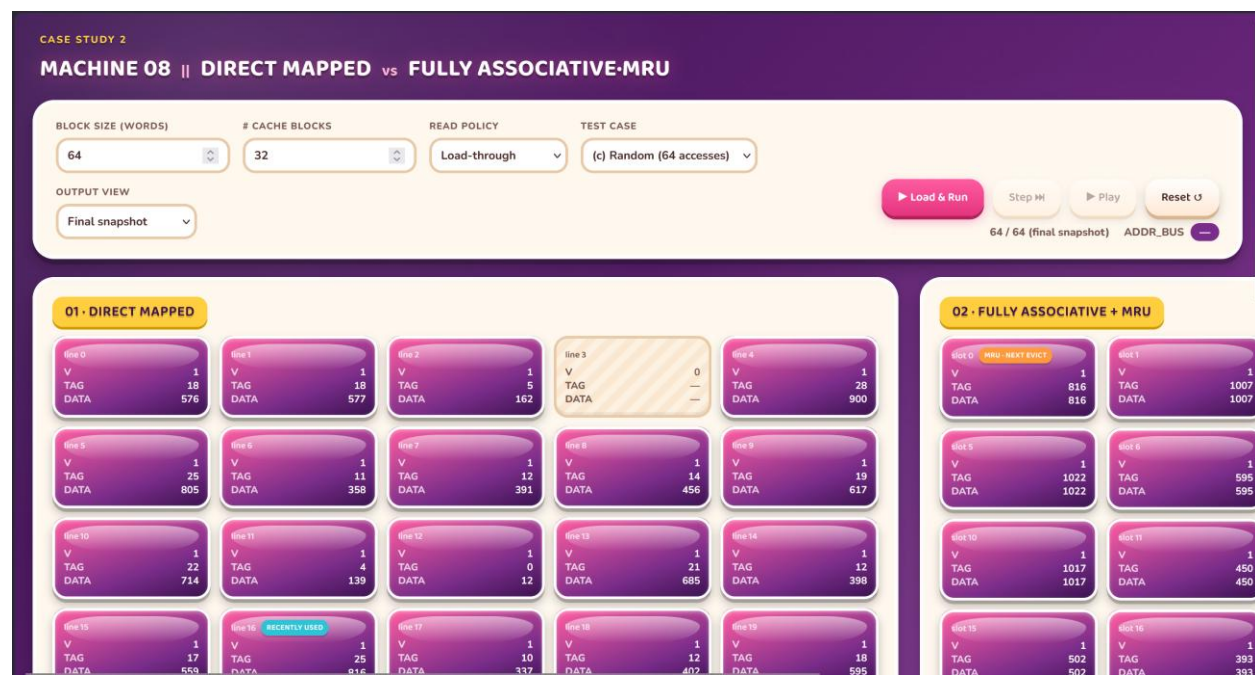
Total accesses	64
Hits	1
Misses	63
Hit rate	1.56%
Miss rate	98.44%
Avg. access time	631.98 cyc
Total access time	40447 cyc

TRACE LOG

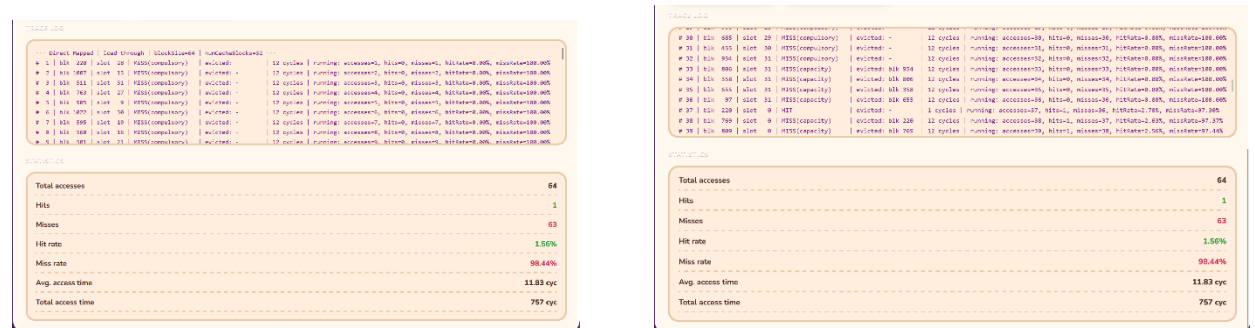
```
--- Fully Associative + MRU | non-load-through | blocksize=64 | numCacheBlocks=32 ---
# 1 | blk 380 | slot 0 | MISS(compulsory) | evicted: - | 642 cycles | running: accesses=1, hits=0, misses=1, hitRate=0.00%, missRate=100.00%
# 2 | blk 930 | slot 1 | MISS(compulsory) | evicted: - | 642 cycles | running: accesses=2, hits=0, misses=2, hitRate=0.00%, missRate=100.00%
# 3 | blk 41 | slot 2 | MISS(compulsory) | evicted: - | 642 cycles | running: accesses=3, hits=0, misses=3, hitRate=0.00%, missRate=100.00%
# 4 | blk 781 | slot 3 | MISS(compulsory) | evicted: - | 642 cycles | running: accesses=4, hits=0, misses=4, hitRate=0.00%, missRate=100.00%
# 5 | blk 838 | slot 4 | MISS(compulsory) | evicted: - | 642 cycles | running: accesses=5, hits=0, misses=5, hitRate=0.00%, missRate=100.00%
# 6 | blk 778 | slot 5 | MISS(compulsory) | evicted: - | 642 cycles | running: accesses=6, hits=0, misses=6, hitRate=0.00%, missRate=100.00%
# 7 | blk 885 | slot 6 | MISS(compulsory) | evicted: - | 642 cycles | running: accesses=7, hits=0, misses=7, hitRate=0.00%, missRate=100.00%
# 8 | blk 622 | slot 7 | MISS(compulsory) | evicted: - | 642 cycles | running: accesses=8, hits=0, misses=8, hitRate=0.00%, missRate=100.00%
# 9 | blk 355 | slot 8 | MISS(compulsory) | evicted: - | 642 cycles | running: accesses=9, hits=0, misses=9, hitRate=0.00%, missRate=100.00%
```

STATISTICS

Total accesses	64
Hits	1
Misses	63
Hit rate	1.56%
Miss rate	98.44%
Avg. access time	631.98 cyc
Total access time	40447 cyc



1st iteration



[illegible][illegible][illegible][illegible][illegible][illegible]

Total accesses	64
Hits	2
Misses	62
Hit rate	3.13%
Miss ratio	96.87%
Avg. access time	11.66 nsec
Total access time	748 cyc

[illegible]